AdaBoost for classification

Decision stumps, exponential loss, weighted vote

Data: twelve training runs (the same as in Part 2)

Summary.
1. Many *very simple* rules, each one only a little better than guessing, are each right about some runs and wrong about others.
2. Give every run an *importance* (a weight). At the start all runs matter equally.
3. After each rule, make the runs it got **wrong** matter **more**, so the next rule is forced to look at them.
4. The final answer is a **vote** of all the rules, where a rule that was right more often gets a louder voice.

Colour key: every symbol keeps its colour in every formula

y	**the truth**	$+1$ if the run diverged, -1 if it converged
h	**the stump's answer**	$+1$ (diverges) or -1 (converges): what one stump says
w	**the weight of a run**	how much the run counts when mistakes are counted; they add to 1
ε	**the weighted error**	total weight of the runs a stump gets wrong
α	**the vote weight**	$\alpha = \frac{1}{2}\ln\frac{1-\varepsilon}{\varepsilon}$: how loud the stump's voice is
Z	**the normaliser**	the number we divide by so the weights add to 1 again
F	**the running vote**	$F = \sum \alpha\, h$: the total of the weighted votes so far
E	**the exponential loss**	$E = \sum e^{-yF}$: the total the committee tries to shrink

Questions a tree can ask: rate batch warm-up ; outcomes: **D** diverged, **C** converged.

Rounds and stumps. One **round** means: pick *one* stump on the current weights, measure its error, work out its vote α, and change the weights. So the number of stumps always equals the number of rounds: after 3 rounds there are 3 stumps. We stop after T rounds; here $T = 6$, so there are six stumps in the committee. (T is our choice, made on separate validation data in practice.) A stump asks a single question, so inside one round there is only one question.

1. The data and the simple rule

A neural network was trained twelve times. For each run we know the learning rate, the batch size, whether warm-up was used, and whether training **D** *diverged* or **C** *converged*. AdaBoost wants labels as numbers: $y = +1$ for **D** and $y = -1$ for **C**.

run	rate	batch	warm-up	outcome	y
1	0.3	256	yes	D	+1
2	0.3	32	no	D	+1
3	0.1	128	yes	D	+1
4	0.1	64	no	D	+1
5	0.03	32	no	D	+1
6	0.01	32	no	D	+1
7	0.03	256	no	C	−1
8	0.03	64	yes	C	−1
9	0.01	128	yes	C	−1
10	0.01	32	yes	C	−1
11	0.001	64	no	C	−1
12	0.001	256	yes	C	−1

The simple rule is a stump: a decision tree with a *single question*, for example "is the rate ≤ 0.065?". It has two leaves, one for yes and one for no, and each leaf answers +1 (D) or −1 (C). From these data there are exactly eight possible questions: four on the rate (thresholds 0.0055, 0.02, 0.065, 0.2), three on the batch size (48, 96, 192) and one on warm-up.

What does each leaf answer? The class that has the larger total *weight* among the runs that land in it. (At the beginning, all weights are equal, so this is just the majority.)

2. Round 1, step by step

1 Step 1. Every run starts with the same weight. $w_i = \dfrac{1}{12} = 0.0833$. The weights always add up to 1. Think of the weight as "how much this run counts when we count mistakes".

2 Step 2. Find the weighted error of every stump. The weighted error ε of a stump is the total weight of the runs it gets *wrong*. Right now this is just "number wrong ÷12".

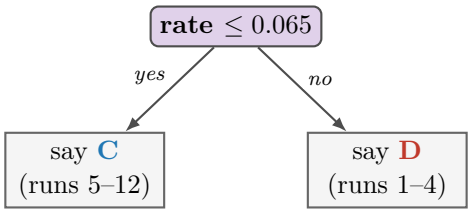
question	yes-leaf says	no-leaf says	runs wrong	ε	
rate ≤ 0.065	C	D	2	0.1667	**smallest**
rate ≤ 0.02	C	D	3	0.2500	
rate ≤ 0.0055	C	D	4	0.3333	
rate ≤ 0.2	C	D	4	0.3333	
batch ≤ 48	D	C	4	0.3333	
warm-up = no	D	C	4	0.3333	
batch ≤ 96	D	C	5	0.4167	
batch ≤ 192	D	C	5	0.4167	

3 Step 3. Keep the stump with the smallest ε. It is the question rate ≤ 0.065: if the rate is small say C, otherwise say D. It is wrong on two runs, **5 and 6**, so $\varepsilon_1 = \frac{2}{12} = \frac{1}{6} = 0.1667$.

Your turn 1: weighted error of another stump

Take the question warm-up = no. Which runs answer yes? What does each leaf say (the class with more runs)? Which runs does the stump get wrong, and what is ε?

Hint: list the runs in each leaf from the table, count D and C in each; the leaf says the larger count; every run on the other side is wrong; ε = number wrong / 12.



wrong on runs 5 and 6: $\varepsilon_1 = 0.1667$

Figure 1. The stump of round 1.

4 Step 4. Decide how loud this stump's voice is. A stump that is almost always right should count a lot; a stump that is barely better than a coin toss should count very little. AdaBoost gives the stump the **vote weight**

$$\boxed{\alpha = \frac{1}{2}\ln\frac{1-\varepsilon}{\varepsilon}} \qquad \text{round 1:} \quad \alpha_1 = \frac{1}{2}\ln\frac{0.8333}{0.1667} = \frac{1}{2}\ln 5 = 0.805.$$

How α behaves (Figure 2): a coin toss $\varepsilon = \frac{1}{2}$ gives $\alpha = 0$, no voice at all. The smaller the error, the larger the voice.

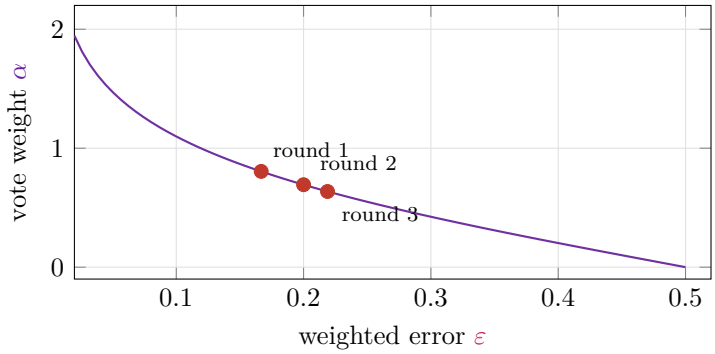


Figure 2. Vote weight against weighted error. The three red points are the first three rounds. At $\varepsilon = 0.5$ the stump is a coin toss and gets no vote.

Your turn 2: the voice of a stump

A stump (not in our problem) has weighted error $\varepsilon = 0.25$. Compute α. Is its voice louder or quieter than the round-1 stump (whose ε was 0.1667)?

Hint: $\frac{1-\varepsilon}{\varepsilon}$ = ?, then ln *of it, then half. Use* $\ln 3 = 1.0986$.

5 Step 5. Change the weights, so that the next stump must look at the mistakes. Every weight is multiplied by a factor, and then everything is divided by their total Z so that the weights

again add up to 1:

$$w_i \leftarrow \frac{w_i\, e^{-\alpha\, y_i\, h(x_i)}}{Z}, \qquad Z = \sum_i w_i\, e^{-\alpha\, y_i\, h(x_i)}.$$

Read the exponent: $y_i h(x_i) = +1$ when the stump is *right* about run i and -1 when it is *wrong*. So

- a run the stump got **right** is multiplied by $e^{-\alpha_1} = 1/\sqrt{5} = 0.447$ (it counts less);
- a run the stump got **wrong** is multiplied by $e^{+\alpha_1} = \sqrt{5} = 2.236$ (it counts more).

run	truth	stump is	w before	factor	w·factor	w after
1	D	right	0.0833	0.447	0.0373	0.0500
2	D	right	0.0833	0.447	0.0373	0.0500
3	D	right	0.0833	0.447	0.0373	0.0500
4	D	right	0.0833	0.447	0.0373	0.0500
5	D	wrong	0.0833	2.236	0.1863	0.2500
6	D	wrong	0.0833	2.236	0.1863	0.2500
7	C	right	0.0833	0.447	0.0373	0.0500
8	C	right	0.0833	0.447	0.0373	0.0500
9	C	right	0.0833	0.447	0.0373	0.0500
10	C	right	0.0833	0.447	0.0373	0.0500
11	C	right	0.0833	0.447	0.0373	0.0500
12	C	right	0.0833	0.447	0.0373	0.0500
			totals: Z_1 and 1		0.7454	1.0000

Your turn 3: re-weight two runs

With $\alpha_1 = 0.805$ ($e^{-\alpha_1} = 0.447$, $e^{\alpha_1} = 2.236$) and $Z_1 = 0.7454$, compute the new weight of run 1 (the stump is right about it) and of run 5 (wrong). Start from $w = 0.0833$ each.

Hint: multiply by the factor first (that is the raw weight), then divide by Z_1.

6 Step 6. What just happened? Runs 5 and 6 jumped from 0.083 to 0.250 each, and every other run fell to 0.050. Together, the two wrong runs now hold weight 0.50: **half of everything**. If we used the same stump again, its weighted error would now be exactly $\frac{1}{2}$, a coin toss, and it would earn no vote. So the next round *has to* find a different question, one that handles runs 5 and 6 better. That is the whole mechanism.

3. Round 2 and round 3: the same steps again

Repeat Steps 2 to 6 with the new weights. In round 2 three questions tie for the smallest weighted error; the tie rule used here and in the deck is: take the feature used the fewest times so far, then the first in the list.

question (round 2)	yes-leaf	no-leaf	ε on the new weights	
rate ≤ 0.0055	C	D	0.2000	
batch ≤ 48	D	C	0.2000	**chosen (tie rule)**
warm-up = no	D	C	0.2000	
batch ≤ 96	D	C	0.2500	
batch ≤ 192	D	C	0.2500	
rate ≤ 0.02	D	D	0.3000	

Your turn 4: round 2, the weighted error

In round 2 runs 5 and 6 weigh 0.25 each and every other run 0.05. For the question batch ≤ 48 (yes: batch 32, i.e. runs 2, 5, 6, 10), say what each leaf answers by total *weight*, then add the weights of the runs it gets wrong.

Hint: yes-leaf: add up the weight of its D runs and of its C runs; the larger wins. Same for the no-leaf. Compare with the table above.

Round 2. The stump is batch ≤ 48 : $\varepsilon_2 = 0.200$, $\alpha_2 = \frac{1}{2}\ln\frac{0.8}{0.2} = \frac{1}{2}\ln 4 = 0.693$. It is wrong on runs 1, 3, 4, 10, so those runs gain weight (the old favourites 5 and 6 are now right, so they lose weight, from 0.250 to 0.156).

Round 3. The new weights are in Figure 3 (middle). The stump is rate ≤ 0.0055 : $\varepsilon_3 = 0.2188$, $\alpha_3 = 0.636$; wrong on runs 7, 8, 9, 10.

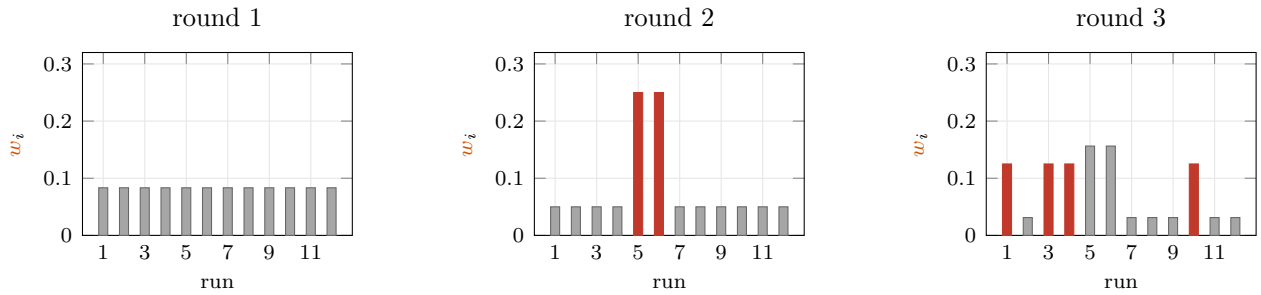


Figure 3. The weights each round starts with. Red: runs the *previous* stump got wrong. Round 1 starts equal. In round 2 runs 5 and 6 are tall; in round 3 the runs round 2 got wrong (1, 3, 4, 10) are tall.

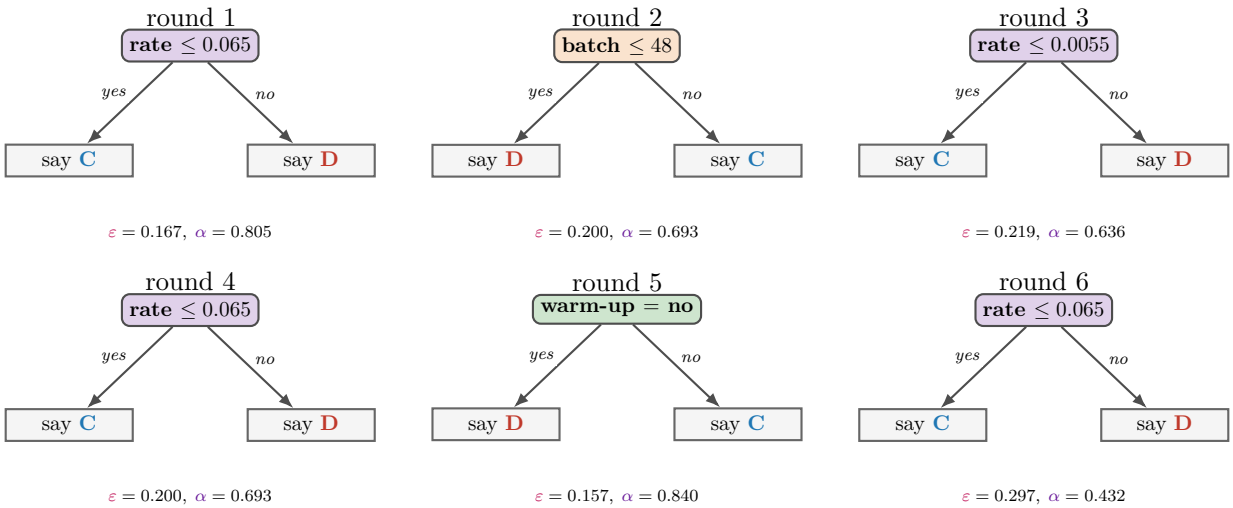


Figure 4. All six stumps, with their weighted error ε and vote weight α. Rounds 1 and 4 and 6 use the same question (rate ≤ 0.065) on different weights.

round	question	yes / no say	ε	α	stump wrong	committee wrong
1	rate ≤ 0.065	C / D	0.1667	0.805	5, 6	5, 6
2	batch ≤ 48	D / C	0.2000	0.693	1, 3, 4, 10	5, 6
3	rate ≤ 0.0055	C / D	0.2188	0.636	7, 8, 9, 10	10
4	rate ≤ 0.065	C / D	0.2000	0.693	5, 6	5, 6
5	warm-up = no	D / C	0.1571	0.840	1, 3, 7, 11	none
6	rate ≤ 0.065	C / D	0.2966	0.432	5, 6	none

A question can come back. Round 4 picks rate ≤ 0.065 again, the very question of round 1, because on the weights of round 4 it is again the best. That is fine: the weights have changed, so it is a different problem.

4. The final answer: a weighted vote

After T rounds we have T stumps $h_1, \dots, h_T$, each with its voice α_k. The prediction for a new run x is

$$H(x) = \operatorname{sign}\Big(\sum_{k=1}^{T} \alpha_k \, h_k(x)\Big)$$

Each stump votes $+1$ (D) or -1 (C); its vote is multiplied by α_k; the votes are added; if the total is positive the answer is D, otherwise C. A louder stump can outvote several quieter ones.

Example: a new run B with rate 0.2, batch 64 and warm-up (it is *not* one of the twelve), six rounds:

round	stump's question	B answers	stump says	α_k	vote $\alpha_k h_k$	running total
1	rate ≤ 0.065	no	D	0.805	+0.805	+0.805
2	batch ≤ 48	no	C	0.693	−0.693	+0.112
3	rate ≤ 0.0055	no	D	0.636	+0.636	+0.748
4	rate ≤ 0.065	no	D	0.693	+0.693	+1.441
5	warm-up = no	no	C	0.840	−0.840	+0.601
6	rate ≤ 0.065	no	D	0.432	+0.432	+1.033

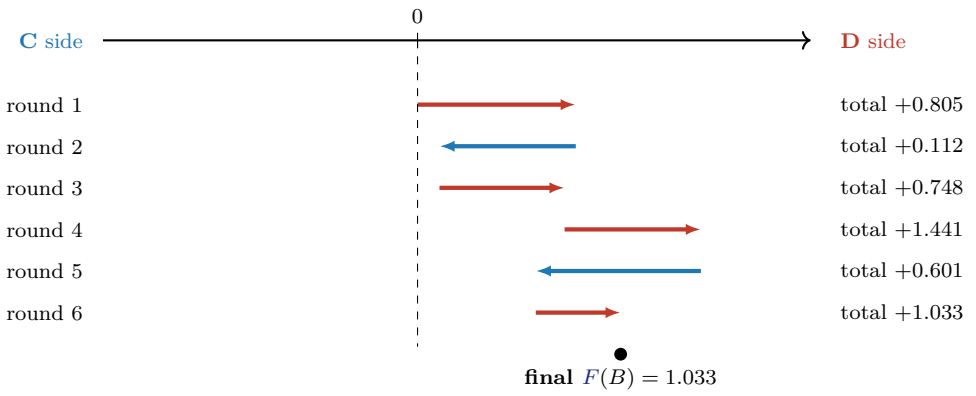


Figure 5. The vote of the six stumps for run B, one arrow per round: red pushes towards D, blue towards C, the length is α_k. The stumps disagree (rounds 2 and 5 vote C), but the total ends clearly on the D side.

The stumps disagree, as a committee does: rounds 2 and 5 say C, the rest say D. The sum is $F(B) = 1.033 > 0$, so AdaBoost answers D. This makes sense: B has a large learning rate.

A harder run. For run A (rate 0.05, batch 32, no warm-up) the six votes add to only 0.240: the answer is still **D**, but only just. A run that looks like runs 5 and 6 (small rate, but small batch and no warm-up) is exactly where a single stump goes wrong, and where the committee is unsure.

How well does the committee do on the twelve training runs? Number of runs it gets wrong after rounds 1 to 6: 2, 2, 1, 2, 0, 0. After round 5 no training run is wrong.

5. Derive it yourself

This section lets you *discover* the two formulas, $\alpha = \frac{1}{2}\ln\frac{1-\varepsilon}{\varepsilon}$ and the re-weighting rule, instead of being told them. We measure how badly the committee does by the **exponential loss**

$$E = \sum_{i=1}^{N} e^{-y_i F(x_i)}, \qquad F = \sum_k \alpha_k h_k \quad \text{(the running total of votes)}.$$

A run the committee gets right ($yF > 0$) contributes a small number; a run it gets wrong ($yF < 0$) contributes a large one. We add one stump at a time and choose the stump h and voice $\alpha > 0$ that make E as small as possible. Work through the boxes in order; each hint is only a nudge, and the full working is in Section 8.

Box D1: split the new loss

Before adding the new stump the committee is F_{old}. After adding it, $F_{\text{new}} = F_{\text{old}} + \alpha h$. Write E_{new} using $w_i = e^{-y_i F_{\text{old}}(x_i)}$ (the loss run i already carries).

Hint: $e^{a+b} = e^a e^b$. Your answer should have the form $\sum_i w_i\, e^{?}$.

Box D2: right and wrong

For one run, $y_i h(x_i)$ is $+1$ if the new stump is right about it and -1 if wrong. Let W_{right} and W_{wrong} be the total w_i of the runs it gets right and wrong. Write E_{new} in terms of W_{right}, W_{wrong} and α.

Hint: split the sum into two groups; in each group the exponent is a single number.

Box D3: which stump?

Take any fixed $\alpha > 0$. Compare e^{α} and $e^{-\alpha}$. Which stump makes E_{new} the smallest: the one with the largest or the smallest W_{wrong}? So how should the stump be chosen?

Hint: a wrong run costs e^{α}, a right run only $e^{-\alpha}$; the total $W_{right} + W_{wrong}$ is fixed.

Box D4: find α

Differentiate $E_{\text{new}} = W_{\text{right}} e^{-\alpha} + W_{\text{wrong}} e^{\alpha}$ with respect to α, set the derivative to zero, and solve for α. Then write $W_{\text{right}}/W_{\text{wrong}}$ using the weighted error $\varepsilon = \dfrac{W_{\text{wrong}}}{W_{\text{right}} + W_{\text{wrong}}}$.

Hint: you will get $e^{2\alpha} = \ldots$; take the logarithm and divide by 2.

Box D5: how small does the loss get?

Put your α back in. Dividing by $W_{\text{right}} + W_{\text{wrong}}$, show that the smallest value is $Z = \varepsilon e^{\alpha} + (1-\varepsilon)e^{-\alpha} = 2\sqrt{\varepsilon(1-\varepsilon)}$. Check with $\varepsilon = \frac{1}{6}$.

Hint: from D4, $e^{\alpha} = \sqrt{(1-\varepsilon)/\varepsilon}$ and $e^{-\alpha} = \sqrt{\varepsilon/(1-\varepsilon)}$.

Box D6: the new weights

After the step, run i carries the loss $e^{-y_i F_{\text{new}}(x_i)}$. Show that it equals $w_i\, e^{-\alpha y_i h(x_i)}$. Why is this exactly the re-weighting rule of Step 5? What is the (unnormalised) total weight after the step?

Hint: use $F_{new} = F_{old} + \alpha h$ again; compare with Step 5.

Box D7: why the committee gets better

(a) A run is classified wrongly only if $y_i F \le 0$. Show that $\mathbf{1}[\text{wrong}] \le e^{-y_i F}$. (b) So the fraction wrong is at most the average of $e^{-y_i F_T}$. Each round multiplies the total loss by Z_k and it starts at N (all $F = 0$). What is the bound? (c) Why does it shrink when every $\varepsilon_k < \frac{1}{2}$?

Hint: (a) $e^{-yF} \ge 1$ when $yF \le 0$. (c) $Z = 2\sqrt{\varepsilon(1-\varepsilon)} < 1$ if $\varepsilon \ne \frac{1}{2}$.

6. The recipe, and common confusions

AdaBoost in five lines.
1. Give every run the weight $1/N$.
2. Pick the stump with the smallest weighted error ε.
3. Its voice is $\alpha = \frac{1}{2}\ln\frac{1-\varepsilon}{\varepsilon}$.
4. Multiply each weight by $e^{-\alpha y h}$ (right runs shrink, wrong runs grow), then divide by the total so the weights add to 1.
5. Repeat, then predict $\operatorname{sign}\sum_k \alpha_k h_k(x)$.

"Why not just use the best stump alone?" It is wrong on two of the twelve runs and cannot be improved by itself: one question cannot separate runs 5 and 6 from the others. Adding stumps lets the committee draw a more complicated boundary out of simple pieces.

"Why do wrong runs get heavier, not lighter?" Because the next stump chooses the question with the smallest *weighted* error. If the wrong runs weigh more, getting them wrong becomes expensive, so the next stump is steered towards them.

"Can the weighted error be above $\frac{1}{2}$?" Not when each leaf says the class with the larger weight: each leaf is then wrong on at most half of its weight, so $\varepsilon \le \frac{1}{2}$ and $\alpha \ge 0$.

"How is it different from bagging?" Bagging grows its trees independently and averages them. AdaBoost grows its stumps one after another, each shaped by the mistakes of the ones before, and the final vote is weighted.

"Where do the other two methods come from?" Gradient boosting and XGBoost keep this same idea but pass the mistakes on differently: not as weights but as residuals (gradient boosting) or as slope and curvature (XGBoost). AdaBoost is gradient boosting for the exponential loss (Section 5, item 4). For *regression* (predicting a number), see the companion notes *AdaBoost for regression*.

7. Check yourself

Fill in the blanks without looking, then check on the next page.

1. The weight of every run at the start is _____.

2. The weighted error of a stump is the total _____ of the runs it gets _____.

3. Round 1: $\varepsilon_1 =$ _____, so $\alpha_1 = \frac{1}{2}\ln$ _____ = _____.

4. A stump with $\varepsilon = \frac{1}{2}$ has $\alpha =$ _____, so it has _____ in the vote.

5. A right run's weight is multiplied by _____, a wrong run's by _____, before dividing by Z.

6. After round 1, runs 5 and 6 each have weight _____, and together they hold _____ of the total weight.

7. So the round-1 stump now has weighted error _____, and round 2 must choose a _____ question.

8. The final answer is $\text{sign}(\sum_k$ _____ $h_k(x))$: a _____ vote.

9. We choose the stump with the smallest weighted error because $E = W_{\text{right}}e^{-\alpha} + W_{\text{wrong}}e^{\alpha}$ and e^{α} is _____ than $e^{-\alpha}$.

10. A run's weight at the start of a round equals its share of the total _____ loss.

Answers

1. $1/12 = 0.0833$

2. weight; wrong

3. $1/6 = 0.1667$; 5; 0.805

4. 0; no say

5. $e^{-\alpha} = 0.447$; $e^{\alpha} = 2.236$

6. 0.250; 0.50

7. 0.5; different

8. α_k; weighted

9. larger (a mistake costs more than a correct run saves)

10. exponential

8. Solutions to the boxes

Your turn 1: the stump warm-up = no

Yes-leaf (no warm-up): runs 2, 4, 5, 6, 7, 11, of which **D**: 2, 4, 5, 6 (four) and **C**: 7, 11 (two). More **D**, so it says **D**. No-leaf (warm-up): runs 1, 3, 8, 9, 10, 12, of which **D**: 1, 3 (two) and **C**: 8, 9, 10, 12 (four). It says **C**. Wrong: runs 7, 11 (said **D**, truly **C**) and 1, 3 (said **C**, truly **D**): four runs, so $\varepsilon = 4/12 = 0.3333$. This is the value in the table of Step 2.

Your turn 2

$\frac{1-0.25}{0.25} = 3$, so $\alpha = \frac{1}{2}\ln 3 = 0.549$. Quieter than 0.805: a larger error earns a smaller voice.

Your turn 3

Run 1 (right): raw $0.0833 \times 0.447 = 0.0373$, new $0.0373/0.7454 = 0.0500$. Run 5 (wrong): raw $0.0833 \times 2.236 = 0.1863$, new $0.1863/0.7454 = 0.2500$. (Check: ten right runs $\times 0.05 = 0.5$, two wrong runs $\times 0.25 = 0.5$; total 1.)

Your turn 4

Yes-leaf (runs 2, 5, 6, 10): **D** weight $0.05 + 0.25 + 0.25 = 0.55$ (runs 2, 5, 6), **C** weight 0.05 (run 10). It says **D**. No-leaf (runs 1, 3, 4, 7, 8, 9, 11, 12): **D** weight 0.15 (runs 1, 3, 4), **C** weight 0.25 (runs 7, 8, 9, 11, 12). It says **C**. Wrong: run 10 (0.05) and runs 1, 3, 4 (0.15): $\varepsilon_2 = 0.20$, matching the table.

Box D1

$E_{\text{new}} = \sum_i e^{-y_i F_{\text{new}}(x_i)} = \sum_i e^{-y_i[F_{\text{old}}(x_i)+\alpha h(x_i)]} = \sum_i e^{-y_i F_{\text{old}}(x_i)}\, e^{-\alpha y_i h(x_i)} = \sum_i w_i\, e^{-\alpha\, y_i h(x_i)}$.

Box D2

If the stump is right on run i, $y_i h(x_i) = +1$ and the factor is $e^{-\alpha}$; if wrong, $y_i h = -1$ and the factor is $e^{+\alpha}$. Collecting the two groups: $E_{\text{new}} = W_{\text{right}}\, e^{-\alpha} + W_{\text{wrong}}\, e^{\alpha}$.

Box D3

For $\alpha > 0$, $e^{\alpha} > e^{-\alpha}$: moving a run from "right" to "wrong" raises its cost from $w\, e^{-\alpha}$ to $w\, e^{\alpha}$. With the total fixed, E_{new} is smallest when W_{wrong} is smallest. So choose the stump with the **smallest weighted error**.

Box D4

$\frac{dE}{d\alpha} = -W_{\text{right}} e^{-\alpha} + W_{\text{wrong}} e^{\alpha} = 0 \;\Rightarrow\; e^{2\alpha} = \frac{W_{\text{right}}}{W_{\text{wrong}}}$. With $\varepsilon = \frac{W_{\text{wrong}}}{W_{\text{right}} + W_{\text{wrong}}}$: $\frac{W_{\text{right}}}{W_{\text{wrong}}} = \frac{1-\varepsilon}{\varepsilon}$. Taking logarithms, $\alpha = \frac{1}{2}\ln\frac{1-\varepsilon}{\varepsilon}$. (Second derivative $W_{\text{right}} e^{-\alpha} + W_{\text{wrong}} e^{\alpha} > 0$: it is a minimum.)

Box D5

From D4, $e^{\alpha} = \sqrt{(1-\varepsilon)/\varepsilon}$ and $e^{-\alpha} = \sqrt{\varepsilon/(1-\varepsilon)}$. Then $Z = \varepsilon\sqrt{\frac{1-\varepsilon}{\varepsilon}} + (1-\varepsilon)\sqrt{\frac{\varepsilon}{1-\varepsilon}} = \sqrt{\varepsilon(1-\varepsilon)} + \sqrt{\varepsilon(1-\varepsilon)} = 2\sqrt{\varepsilon(1-\varepsilon)}$. For $\varepsilon = \frac{1}{6}$: $2\sqrt{\frac{5}{36}} = 0.7454$, which is the Z_1 of the weights table.

Box D6

$e^{-y_i F_{\text{new}}} = e^{-y_i F_{\text{old}}}\, e^{-\alpha y_i h(x_i)} = w_i\, e^{-\alpha y_i h(x_i)}$. This is the numerator of the re-weighting rule: right runs ($yh = +1$) are multiplied by $e^{-\alpha}$, wrong runs by e^{α}. The total of these new numbers is $E_{\text{new}} = Z \times (W_{\text{right}} + W_{\text{wrong}})$; dividing by it is the "$\div Z$" that makes the weights add to 1. *A run's weight is its share of the total exponential loss.*

Box D7

(a) If $y_i F \le 0$ the run is wrongly classified and $e^{-y_i F} \ge 1$. If $y_i F > 0$ the run is right and $e^{-y_i F} > 0 = \mathbf{1}[\text{wrong}]$. Either way $\mathbf{1}[\text{wrong}] \le e^{-y_i F}$. (b) Hence $\dfrac{\#\text{wrong}}{N} \le \dfrac{1}{N}\sum_i e^{-y_i F_T(x_i)}$. At the start $F = 0$, so the total loss is N. Round k multiplies the total by Z_k (Box D6), so the average loss after T rounds is $Z_1 Z_2 \cdots Z_T$, and this is the bound. (c) $Z = 2\sqrt{\varepsilon(1-\varepsilon)}$ is below 1 unless $\varepsilon = \frac{1}{2}$, so every round with $\varepsilon < \frac{1}{2}$ multiplies the bound by a number below 1. Numbers: $Z_1 = 0.7454$, $Z_2 = 0.8000$, $Z_3 = 0.8268$, product 0.4930 = the average exponential loss after round 3, computed directly from the scores.